

One-Bit Consensus Control of Multi-Agent Systems With Packet Loss

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Abstract—This letter investigates the one-bit consensus control of multi-agent systems (MASs) with independent and identically distributed and Markovian packet loss. To explore the impact of packet loss on one-bit communication, this letter first quantitatively characterizes the information loss of one-bit communications caused by packet loss, which provides the proportional relationship between one-bit data with and without packet loss in the sense of expectation. Based on quantitative characterizations, a one-bit packet loss consensus algorithm with a packet loss proportional coefficient is proposed to compensate for the information loss, where the coefficient is designed as the reciprocal of the information loss proportion. Furthermore, this letter demonstrates that the proposed algorithm enables the MAS to achieve one-bit consensus in the mean square sense at a rate of $O(1/t)$ with packet loss. Two simulation examples are given to validate the algorithm.

Index Terms—Multi-agent systems, one-bit consensus, packet loss.

I. INTRODUCTION

OVER decades, consensus problems in multi-agent systems (MASs) over digital networks has become a central topic in distributed control [1], [2], [3], [4], driven by

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the rapid advancements in digital communication technologies. Among the various approaches to overcome the limited capacity of digital networks, one-bit consensus has gained attention for its ability to achieve consensus with only one bit of communication per unit of time, which can reduce communication cost compared to conventional methods. As a result, there has been considerable research on one-bit consensus in the past few years [5], [6], [7], [8]. Specifically, for the first-order MAS, Zhao et al. [5] constructed a two-time-scale consensus algorithm to make MASs achieve mean square consensus; Wang et al. [6] proposed a consensus algorithm based on a recursive projection algorithm and obtained a faster consensus rate than that given in [5]. For the high-order MAS, a compression-control type one-bit consensus algorithm was proposed to achieve a one-bit consensus for controllable MASs in [8].

In digital networks, some communication information among agents is often lost due to network congestion, channel interference, cyber attacks, and other problems, a process known as packet loss [9], [10], [11]. Consequently, numerous studies have investigated consensus problems with packet loss [12], [13], [14], [15], [16], which is typically modeled as an independent and identically distributed (i.i.d.) Bernoulli process or a more general Markov process. For i.i.d. Bernoulli packet loss, consensus conditions and rates were analyzed in [12], while sufficient conditions for consensus in linear discrete-time agents were derived in [13] and [14]. For Markovian packet loss, analytical consensus conditions under Markovian packet loss were established in [15], while Ballam et al. [16] proposed a compensation mechanism to estimate a neighbor's information when packet loss occurs. However, the above studies on consensus with packet loss consider only ideal communication without bandwidth limitations, which does not align with practical digital networks.

This letter investigates one-bit consensus of MASs with i.i.d. and Markovian packet loss. Inspired by our previous one-bit consensus work [8] without packet loss, the impact of packet loss on one-bit consensus is explored to overcome the resulting challenge. The main contributions are as follows:

- This letter provides quantitative characterizations of the information loss of one-bit communication data caused by i.i.d. and Markovian packet loss. Specifically, proportional relationships between one-bit data with and without packet loss in the sense of expectation are established. The finding of these quantitative characterizations

facilitates the design of the consensus algorithm, which can effectively deal with packet loss.

- A one-bit packet loss consensus algorithm is proposed, which introduces a packet loss proportional coefficient to compensate for information loss. Based on the quantitative characterizations, this letter designs the packet loss proportional coefficients as the reciprocal of information loss proportions for both i.i.d. and Markovian packet loss. Furthermore, this letter demonstrates that the proposed algorithm enables MAS to achieve one-bit consensus in the mean square sense at a rate of $O(\frac{1}{t})$ with packet loss.

The remainder of this letter is organized as follows. Section II provides the preliminaries and describes the consensus problem. Section III introduces the consensus algorithm. The main results are presented in Section IV, which includes consensus results with both i.i.d. and Markovian packet loss. Section V gives two simulation examples. Section VI summarizes the findings and outlines future directions.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Preliminaries

To describe the relation between agents, this letter introduces a topology $\mathcal{G} = (\mathcal{N}_0, \mathcal{E})$, where $\mathcal{N}_0 = \{1, \dots, N\}$ is the set of agents, and $\mathcal{E} \subseteq \mathcal{N}_0 \times \mathcal{N}_0$ is the ordered edges set of the topology $\mathcal{G}$. Denote $\mathcal{N}_i$ as the neighbor set of the agent i in the topology $\mathcal{G}$. Denote the adjacency matrix of the N agents as $\mathcal{A}_{\mathcal{G}}$, where each element of the matrix $\mathcal{A}_{\mathcal{G}}$ satisfies $a_{ij} = a_{ji} = 1$ if $(i, j) \in \mathcal{E}$, else $a_{ij} = 0$. Denote the degree matrix of the N agents as $\mathcal{D}$, where $\mathcal{D} = \text{diag}\{d_1, d_2, \dots, d_N\}$ and d_i is the degree of agent i . Denote $d = \sum_{i=1}^N d_i$. Then, the Laplacian matrix of $\mathcal{G}$ is defined as $\mathcal{L} = \mathcal{D} - \mathcal{A}_{\mathcal{G}}$.

B. Problem Formulation

Consider the following MAS with N agents at time t :

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \quad i = 1, \dots, N, \quad (1)$$

where $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^n$ are constant matrices, $x_i(t) \in \mathbb{R}^n$ is the state of the agent i at time t , and $u_i(t) \in \mathbb{R}$ is the control input of the agent i at time t .

Agent i receives one-bit information from its neighbor j , which is affected by communication noise and packet loss:

$$\begin{cases} s_{ij}(t) = \mathbb{1}_{\{g(x_j(t)) + d_{ij}(t) \leq c_{ij}\}}, \\ \tilde{s}_{ij}(t) = \gamma_{ij}(t)s_{ij}(t), \end{cases} \quad (2)$$

where the agent $j \in \mathcal{N}_i$, $g(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}$ is a compression encoding function to be designed, $d_{ij}(t) \in \mathbb{R}$ is the communicating noise, $c_{ij} \in \mathbb{R}$ is the threshold value, $\mathbb{1}_{\{a \leq c\}}$ is an indicative function defined as $\mathbb{1}_{\{a \leq c\}} = 1$ when $a \leq c$ and 0 otherwise, $s_{ij}(t)$ is the one-bit data without packet loss.

$\gamma_{ij}(t)$ models the packet loss of the communication channel (j, i) , which satisfies that $\gamma_{ij}(t) = 1$ when the transmission is successful at time t , and 0 otherwise. $\tilde{s}_{ij}(t)$ is the one-bit data with packet loss that the agent i actually receives from its neighbor j .

To proceed with our analysis, we introduce these assumptions about the system model, the communication noises, and the communication topology.

Assumption 1: The linear system (A, B) is controllable.

Assumption 2: The noises $\{d_{ij}(t), i, j \in \mathcal{N}_0, t \in \mathbb{N}\}$ are i.i.d. as $\mathcal{N}(0, \delta^2)$ for indices i, j and time t , with known distribution function $F(\cdot)$ and density function $f(\cdot)$.

Assumption 3: The topology graph $\mathcal{G}$ is connected.

Problem: The goal of this letter is to design a controller $u_i(t)$ and the communication mechanism $g(\cdot)$ based on one-bit communications $\tilde{s}_{ij}(t)$ with packet loss to achieve consensus of the controllable MAS (1)-(2), i.e., $E[\|x_i(t)\|^2] < \infty$ and $\lim_{t \rightarrow \infty} E[\|x_i(t) - \bar{x}(t)\|^2] = 0$ for all $i = 1, \dots, N$, where $\bar{x}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t)$.

III. ALGORITHM DESIGN

This section focuses on designing a consensus algorithm that can deal with the information loss caused by packet loss and enable MAS to achieve one-bit consensus.

Since any controllable system (1) can be transformed into the Brunovsky canonical form [17], this letter considers the MAS (1)-(2) in the Brunovsky canonical form to simplify the design of the algorithm, where

$$A \triangleq \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \end{bmatrix} \in \mathbb{R}^{n \times n}, B \triangleq \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^n. \quad (3)$$

To compensate for the information loss caused by packet loss, we propose a one-bit packet loss consensus algorithm by incorporating a packet loss proportional coefficient into the compression-control-type one-bit consensus algorithm in our previous work [8]. The algorithm is as follows.

Remark 1: Compared to the compression-control type one-bit consensus algorithm in [8], Algorithm 1 in this letter introduces a packet loss proportional coefficient κ in the estimation procedure (4), which is used to deal with packet loss. Specifically, we aim to compensate for the information loss caused by packet loss by designing the proportional coefficient κ based on the property of packet loss, thereby recovering the one-bit data $s_{ij}(t)$ without packet loss.

The one-bit packet loss consensus control algorithm comprises a communication protocol and a consensus controller. In the communication protocol, the compression encoding function $g(\cdot)$ is introduced to achieve one-bit communication of high-order systems. In the consensus controller, the estimation is used to estimate the compressed state $K_2 x_j(t)$ of the agent from one-bit communications $\tilde{s}_{ij}(t)$. The controller with a stabilization term and a consensus term is proposed to ensure the consensus of a potentially unstable but controllable MAS, where the consensus term is designed based on the above estimate of the compressed state.

Remark 2: Algorithm 1 can be extended to general controllable linear systems (1) by transforming the control gain K_1 and compression coefficient K_2 to $K_1 P$ and $K_2 P$, where P is the transformation matrix that brings the general controllable system to the Brunovsky canonical form. Detailed explanations can be found in Remarks 3 and 7 of [8].

For the convenience of the subsequent analysis, we rewrite the estimation and state update formulas in vector form.

Firstly, denote $x(t) = [x_1^T(t), x_2^T(t), \dots, x_N^T(t)]^T \in \mathbb{R}^{nN}$ and $\hat{z}(t) = [\hat{z}_{1r_1}(t), \dots, \hat{z}_{1r_{d_1}}(t), \dots, \hat{z}_{ir_{d_1+\dots+d_{i-1}+1}}(t), \dots, \hat{z}_{ir_{d_1+\dots+d_i}}(t), \dots, \hat{z}_{Nr_{d_1+\dots+d_N}}(t)]^T \in \mathbb{R}^d$, where $r_{d_1+\dots+d_{i-1}+1}, \dots, r_{d_1+\dots+d_i} \in \mathcal{N}_i$ for $i = 1, \dots, N$. Similarly,

Algorithm 1 One-Bit Packet Loss Consensus Control

Initiation: Denote the integer $t_0(> 0)$ as the initial time. $x_i(t_0 + 1) = x_i^0$ is the initial state of the agent i , $\hat{z}_{ij}(t_0) = \hat{z}_{ij}^0$ is the initial estimate of the agent j estimated by the agent i . Then, for $t \geq t_0 + 1$, the algorithm is as follows.

Step 1: Communication protocol:

Denote the compression encoding function

$$g(x_j(t)) = K_2 x_j(t),$$

where $K_2 = [b_1, \dots, b_{n-1}, 1] \in \mathbb{R}^{1 \times n}$ is the compression coefficient and $b_1, \dots, b_{n-1}$ satisfy that: All the roots $r_1, \dots, r_{n-1}$ of $z^{n-1} + b_{n-1}z^{n-2} + \dots + b_2z + b_1 = 0$ are inside the unit circle.

Step 2: Consensus controller:

Step 2.1 Estimation: each agent i estimates the compressed state $K_2 x_j(t)$ of its neighbor agent j at time t by

$$\hat{z}_{ij}(t) = \Pi_M \left\{ \hat{z}_{ij}(t-1) + \frac{\beta}{t} \left(F_{c_{ij}}(\hat{z}_{ij}(t-1)) - \kappa \tilde{s}_{ij}(t) \right) \right\}, \quad (4)$$

where $j \in \mathcal{N}_i$, $\beta > 0$ is the step coefficient for estimation updating, $F_{c_{ij}}(z) = F(c_{ij} - z)$ for any $z \in \mathbb{R}$, κ is the packet loss proportional coefficient to be designed, $\Pi_M(\zeta) = \arg \min_{|\xi| \leq M} |\zeta - \xi|$ is a projection mapping for any $\zeta \in \mathbb{R}$, where M is the upper bound of $|K_2 x_i^0|$ and $|\hat{z}_{ij}^0|$.

Step 2.2 Controller: based on these estimates, each agent i designs its control by

$$u_i(t) = K_1 x_i(t) + \frac{\gamma}{t+1} \sum_{j \in \mathcal{N}_i} (\hat{z}_{ij}(t) - K_2 x_i(t)), \quad (5)$$

where $K_1 = [-a_1 + b_1, -a_2 + b_2 - b_1, \dots, -a_{n-1} + b_{n-1} - b_{n-2}, -a_n + b_{n-1} + 1] \in \mathbb{R}^{1 \times n}$ is the control gain of the stabilization term $K_1 x_i(t)$, $\gamma > 0$ is the step coefficient of the consensus term $\frac{\gamma}{t+1} \sum_{j \in \mathcal{N}_i} (\hat{z}_{ij}(t) - K_2 x_i(t))$.

denote $\tilde{s}(t) = [\tilde{s}_{1r_1}(t), \dots, \tilde{s}_{Nr_{d_1+\dots+d_N}}(t)]^T \in \mathbb{R}^d$, and $C = [c_{1r_1}, \dots, c_{Nr_{d_1+\dots+d_N}}]^T \in \mathbb{R}^d$.

Then, we construct matrices to establish the relation between agents' compressed states and their estimates. Without loss of generality, assume subscript r_s in vector $\hat{z}(t)$ represents neighbor j of agent i , i.e., $\hat{z}_{ir_s}(t) = \hat{z}_{ij}(t)$, where $r_s \in \mathcal{N}_i$, $s \in \{d_1 + d_2 + \dots + d_{i-1} + 1, \dots, d_1 + \dots + d_i\}$.

Q is designed to select the agent that correlates with its estimate of the compressed state. Define $Q = [Q_{1r_1}, \dots, Q_{Nr_{d_1+\dots+d_N}}]^T \in \mathbb{R}^{d \times N}$, where $Q_{ir_s} = Q_{ij} = [\bar{0}_{j-1}^T, 1, \bar{0}_{N-j}^T]^T \in \mathbb{R}^d$ for $(i, r_s) \in \mathcal{E}$, else $Q_{ir_s} = \bar{0}_N$, $\bar{0}_m = [0, \dots, 0]^T \in \mathbb{R}^m$.

W is designed to select the neighbor set of each agent. Define $W = [W_1, \dots, W_N]^T \in \mathbb{R}^{N \times d}$, where $W_i = [\bar{0}_{d_1+\dots+d_{i-1}}^T, \bar{1}_{d_i}^T, \bar{0}_{d_{i+1}+\dots+d_N}^T]^T \in \mathbb{R}^d$ for $i \in \{1, \dots, N\}$, $\bar{1}_m = [1, \dots, 1]^T \in \mathbb{R}^m$.

Based on the above matrices, the vector forms of estimation and update are given as follows:

1. Estimation:

$$\hat{z}(t) = \Pi_M \left\{ \hat{z}(t-1) + \frac{\beta}{t} (\mathcal{F}(\hat{z}(t-1)) - \kappa \tilde{s}(t)) \right\},$$

where $\Pi_M(z) = [\Pi_M(z_1), \dots, \Pi_M(z_d)]^T$, $\mathcal{F}(z) = [F_{c_1}(z_1), \dots, F_{c_d}(z_d)]^T$, for any $z = [z_1, z_2, \dots, z_d]^T \in \mathbb{R}^d$, c_i is the i th element of threshold vector C .

2. Update:

$$x(t) = \left(I_N \otimes (A + BK_1) - \frac{\gamma}{t} \mathcal{L} \otimes BK_2 \right) x(t-1) + \frac{\gamma}{t} (W \otimes B) \hat{e}(t),$$

where $\hat{e}(t) = \hat{z}(t) - (Q \otimes K_2)x(t)$ is the estimation error of the compressed state $(I_N \otimes K_2)x(t)$.

IV. MAIN RESULTS

This section separately provides a quantitative characterization of the information loss of one-bit communication caused by i.i.d. and Markovian packet loss. Based on this quantitative characterization, the packet loss proportional coefficient κ is designed, thereby recovering the one-bit data without packet loss in the sense of expectation. Furthermore, we demonstrate that all agents can reach the one-bit consensus and provide the corresponding consensus rate.

A. The Case of i.i.d. Packet Loss

In this subsection, we study one-bit consensus with i.i.d. packet loss, focusing on how to mitigate the impact of i.i.d. packet loss on one-bit communications. For the i.i.d. packet loss process, we make the following assumption.

Assumption 4: For all indices i and j , the packet loss sequence $\{\gamma_{ij}(t)\}_{t \in \mathbb{N}}$ is an i.i.d. Bernoulli process with packet loss probability $p \in (0, 1)$, i.e., $P\{\gamma_{ij}(t) = 1\} = 1 - p$ and $P\{\gamma_{ij}(t) = 0\} = p$.

To describe the information loss caused by i.i.d. packet loss, we give the following proposition.

Proposition 1: Under Assumption 4, it holds that

$$E[\tilde{s}_{ij}(t) - (1-p)s_{ij}(t)] = 0,$$

where $(i, j) \in \mathcal{E}$.

Proof: Since the loss effect $\gamma_{ij}(t)$ is independent of the one-bit communication $s_{ij}(t)$, then $E[\tilde{s}_{ij}(t)] = E[\gamma_{ij}(t)s_{ij}(t)] = E[\gamma_{ij}(t)]E[s_{ij}(t)]$. Therefore, we have

$$\begin{aligned} E[\tilde{s}_{ij}(t) - (1-p)s_{ij}(t)] &= E[\gamma_{ij}(t)s_{ij}(t)] - E[(1-p)s_{ij}(t)] \\ &= (1-p)E[s_{ij}(t)] - E[(1-p)s_{ij}(t)] = 0. \end{aligned}$$

Remark 3: Proposition 1 provides a quantitative characterization of the information loss of one-bit data caused by i.i.d. packet loss, i.e., the information is reduced to $(1-p)$ times that without packet loss in the sense of expectation.

Noting that the classical estimation-based consensus algorithms without packet loss in [6], [7], [8] and [18] adopt the term $F_{c_{ij}}(\hat{z}_{ij}(t)) - s_{ij}(t)$ to provide a nonlinear mapping of estimation error in the sense of expectation, indicating the direction of estimation based on one-bit data. This is because $E[F_{c_{ij}}(\hat{z}_{ij}(t)) - s_{ij}(t)] = F_{c_{ij}}(\hat{z}_{ij}(t)) - F_{c_{ij}}(K_2 x_j(t))$ has the opposite sign of the estimation error $K_2 x_j(t) - \hat{z}_{ij}(t)$. Thus, it suffices to recover the one-bit data without packet loss in the expectation sense to ensure the correct estimation direction under packet loss.

Based on the quantitative characterization of information loss in Proposition 1, the packet loss proportional coefficient in one-bit packet loss consensus algorithm is designed as

$$\kappa = \frac{1}{1-p}.$$

Then, the following theorem establishes the consensus properties of MAS with i.i.d. packet loss.

Theorem 1: For the case of i.i.d. packet loss, consider MAS consisting of the plant (1)–(3) and one-bit packet loss consensus algorithm with $\kappa = \frac{1}{1-p}$. If Assumptions 1–4 hold, then the following conclusions hold:

- 1) The MAS reaches consensus, i.e.,

$$\lim_{t \rightarrow \infty} E[\|x_i(t) - \bar{x}(t)\|^2] = 0,$$

providing that $\beta > \frac{1}{2f_M}(\frac{\gamma\lambda_G^2}{\lambda_2^3} + \gamma\alpha)$;

- 2) The MAS reaches consensus at the rate of $O(\frac{1}{t})$, i.e.,

$$E[\|x_i(t) - \bar{x}(t)\|^2] = O\left(\frac{1}{t}\right),$$

if $\beta > \frac{1}{2f_M}(\frac{\gamma\lambda_G^2}{\lambda_2^2(\gamma\lambda_2-1)} + \gamma\alpha + 1)$ and $\gamma > \frac{1}{\lambda_2}$;

where $f_M = \min_{i,j \in \mathcal{N}_0} \{f(|c_{ij}| + M)\}$, λ_2 and λ_G are the smallest positive eigenvalue and the largest eigenvalue of $\mathcal{L}$, $\alpha = 2\sqrt{\lambda_{QW}} + \lambda_{Q\mathcal{L}}\lambda_2/\lambda_G$, $\lambda_{QW} = \|QW\|^2$, $\lambda_{Q\mathcal{L}} = \|Q\mathcal{L}T_G\|^2$, and T_G is the standard eigenvector matrix of $\mathcal{L}$.

Proof: Two Lyapunov functions are introduced to analyze the consensus of the compressed states and the convergence of their estimates, which are

$$V(t) = E[\|(T_G^{-1} \otimes K_2)\delta(t)\|^2], R(t) = E[\|\hat{\varepsilon}(t)\|^2],$$

where $\delta(t) = (J_N \otimes I_n)x(t)$ and $J_N = I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$.

By the definition of $\hat{\varepsilon}(t)$, and the property of Π_M that $|\Pi_M(x_1) - \Pi_M(x_2)| \leq |x_1 - x_2|$, we have

$$\begin{aligned} R(t) &\leq E\left[\|\hat{z}(t-1) + \frac{\beta}{t}\left(\mathcal{F}_C(\hat{z}(t-1)) - \frac{1}{1-p}\tilde{s}(t)\right) - (Q \otimes K_2)x(t)\|^2\right] \\ &= E\left[\hat{\varepsilon}^T(t-1)\left(I_d - \frac{\gamma}{t}QW\right)^T\left(I_d - \frac{\gamma}{t}QW\right)\hat{\varepsilon}(t-1) + \frac{2\gamma}{t}E\left[\hat{\varepsilon}^T(t-1)\left(I_d - \frac{\gamma}{t}QW\right)^T Q\mathcal{L}T_G\hat{\delta}(t-1)\right] + \frac{2\beta}{t}E\left[\hat{\varepsilon}^T(t-1)\left(I_d - \frac{\gamma}{t}QW\right)^T\left(\mathcal{F}_C(\hat{z}(t-1)) - \frac{1}{1-p}\tilde{s}(t)\right)\right] + O\left(\frac{1}{t^2}\right)\right]. \end{aligned} \quad (6)$$

Denote the first, second, and third items of (6) as $R_1(t)$, $R_2(t)$, and $R_3(t)$, respectively, i.e., $R(t) \leq R_1(t) + R_2(t) + R_3(t) + O(\frac{1}{t^2})$. Then, we have

$$R_1(t) \leq \left(1 + \frac{\gamma\sqrt{\lambda_{QW}}}{t}\right)^2 R(t-1). \quad (7)$$

Using the Cauchy-Schwarz inequality, gives

$$\begin{aligned} R_2(t) &\leq \frac{2\gamma}{t} \left(E\left[\hat{\varepsilon}^T(t-1)\left(I_d - \frac{\gamma}{t}QW\right)^T\left(I_d - \frac{\gamma}{t}QW\right)\hat{\varepsilon}(t-1)\right] E\left[\hat{\delta}^T(t-1)T_G^T Q^T Q\mathcal{L}T_G\hat{\delta}(t-1)\right]\right)^{\frac{1}{2}} \\ &\leq \frac{\gamma}{t} \left(\frac{\lambda_{Q\mathcal{L}}\lambda_2}{\lambda_G}(1 + \frac{\gamma\sqrt{\lambda_{QW}}}{t})^2 R(t-1) + \frac{\lambda_G}{\lambda_2} V(t-1)\right). \end{aligned} \quad (8)$$

Moreover, by the communication protocol designed in Algorithm 1 and Proposition 1, it can be seen that

$$E\left[\frac{1}{1-p}\tilde{s}(t)\right] = E[s(t)] = \mathcal{F}_C((Q \otimes K_2)x(t)).$$

By Lagrange's Mean Value Theorem, we have $\mathcal{F}_C(\hat{z}(t-1)) - \mathcal{F}_C((Q \otimes K_2)x(t)) = \text{diag}(\tilde{f}(\zeta(t))) (\hat{z}(t-1) - (Q \otimes K_2)x(t))$, where $\tilde{f} = [f(\cdot), \dots, f(\cdot)] \in \mathbb{R}^d$ and $\zeta(t)$ is between $\hat{z}(t-1)$ and $(Q \otimes K_2)x(t)$.

Similarly to [8, Lemma 1], it can be derived that the compressed states and their estimates are bounded, i.e., $|K_2 x_j(t)| \leq M$ and $|\hat{z}_{ij}(t)| \leq M$, which implies $\text{diag}(\tilde{f}(\zeta(t))) \geq f_M \cdot I_d$. Then,

$$\begin{aligned} R_3(t) &= -\frac{2\beta}{t} E[\hat{\varepsilon}^T(t-1) \text{diag}(\tilde{f}(\zeta(t))) \hat{\varepsilon}(t-1)] + O\left(\frac{1}{t^2}\right) \\ &\leq -\frac{2\beta f_M}{t} R(t-1) + O\left(\frac{1}{t^2}\right). \end{aligned} \quad (9)$$

Taking (7)–(9) into (6) gives

$$\begin{aligned} R(t) &\leq \left(1 - \frac{2\beta f_M - \gamma\alpha}{t}\right) R(t-1) + \frac{\gamma\lambda_G/\lambda_2}{t} V(t-1) \\ &\quad + O\left(\frac{1}{t^2}\right). \end{aligned}$$

Similarly to the analysis of $R(t)$ and [8, Lemma 1], we obtain

$$V(t) \leq \left(1 - \frac{\gamma\lambda_2}{t}\right) V(t-1) + \frac{\gamma\lambda_G/\lambda_2}{t} R(t-1) + O\left(\frac{1}{t^2}\right).$$

The subsequent proof follows a similar approach to that in [8]. Due to space limitations, only an outline is provided.

First, we denote $Z(t) = (V(t), R(t))^T$ to jointly analyze the convergence properties of $V(t)$ and $R(t)$. From these properties, we derive the convergence properties of the estimates and the consensus of the compressed states, which is similar to [8, Th. 1].

Then, based on the equivalence between the consensus of original states and the compressed states, i.e., [8, Lemmas 6–7], we can prove that the MAS can achieve one-bit consensus in the mean square sense at a rate of $O(\frac{1}{t})$ with appropriately chosen step coefficients β and γ . ■

B. The Case of Markovian Packet Loss

This subsection studies one-bit consensus with a more general Markovian packet loss and makes the following assumption.

Assumption 5: For all indices i and j , the packet loss sequence $\{\gamma_{ij}(t)\}_{t \in \mathbb{N}}$ is an ergodic Markov process with two states $\{0, 1\}$, transition probability $p_{uv} = P\{\gamma_{ij}(t) = v - 1 | \gamma_{ij}(t-1) = u - 1\}$, and stationary distribution $\pi_u = \lim_{t \rightarrow \infty} P\{\gamma_{ij}(t) = u - 1\}$, for all $u, v \in \{1, 2\}$.

To describe the information loss caused by Markovian packet loss, we give the following proposition.

Proposition 2: Under Assumption 5, it holds that

$$E[\tilde{s}_{ij}(t) - \pi_2 s_{ij}(t)] = O(\lambda_p^t),$$

where $0 < \lambda_p < 1$, $(i, j) \in \mathcal{E}$.

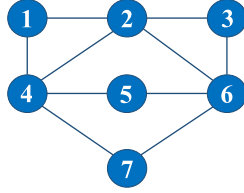


Fig. 1. The fixed communication topology of the multi-agent system.

Proof: Denote $p_{u,t} = P\{\gamma(t) = u - 1\}$, $u = 1, 2$. By [8, Lemma 8], we have $p_{u,t} = \pi_u + O(\lambda_p^t)$, $0 < \lambda_p < 1$. Then, the expectation of $\gamma_{ij}(t)$ can be calculated as

$$\begin{aligned} E[\gamma_{ij}(t)] &= P\{\gamma_{ij}(t) = 1\} \times 1 + P\{\gamma_{ij}(t) = 0\} \times 0 \\ &= p_{2,t} \times 1 + p_{1,t} \times 0 \\ &= \pi_2 + O(\lambda_p^t). \end{aligned}$$

Moreover, it can be seen that

$$\begin{aligned} E[\tilde{s}_{ij}(t) - \pi_2 s_{ij}(t)] &= E[\gamma_{ij}(t) s_{ij}(t) - \pi_2 s_{ij}(t)] \\ &= (\pi_2 + O(\lambda_p^t)) E[s_{ij}(t)] - \pi_2 E[s_{ij}(t)] \\ &= O(\lambda_p^t). \end{aligned}$$

Remark 4: Proposition 2 provides an approximate quantitative characterization of the information loss of one-bit communication caused by Markovian packet loss, with an exponentially decaying error of $O(\lambda_p^t)$. Specifically, in the sense of expectation, the information is approximately reduced to π_2 times that without the packet loss.

Based on the approximate quantitative characterization in Proposition 2, the packet loss proportional coefficient in one-bit packet loss consensus algorithm is designed as

$$\kappa = \frac{1}{\pi_2}.$$

Then, the following theorem establishes the consensus properties of MAS with Markovian packet loss.

Theorem 2: For the case of Markovian packet loss, consider MAS consisting of the plant (1)-(3) and one-bit packet loss consensus algorithm with $\kappa = \frac{1}{\pi_2}$. If Assumptions 1-3 and 5 hold, then the following conclusions hold:

- 1) The MAS reaches consensus, if $\beta > \frac{1}{2f_M} (\frac{\gamma^2 \lambda_G^2}{\lambda_2^3} + \gamma \alpha)$;
- 2) The MAS reaches consensus at the rate of $O(\frac{1}{t})$, under $\beta > \frac{1}{2f_M} (\frac{\gamma^2 \lambda_G^2}{\lambda_2^3 (\gamma \lambda_2 - 1)} + \gamma \alpha + 1)$ and $\gamma > \frac{1}{\lambda_2}$.

Proof: Following the proof of Theorem 1, we get

$$\begin{aligned} R(t) &= E \left[\left\| \Pi_M \left\{ \hat{z}(t-1) + \frac{\beta}{t} \left(\mathcal{F}_C(\hat{z}(t-1)) - \frac{1}{\pi_2} \tilde{s}(t) \right) \right\} \right. \right. \\ &\quad \left. \left. - (Q \otimes K_2)x(t) \right\|^2 \right] \\ &\triangleq R_1(t) + R_2(t) + \bar{R}_3(t) + O\left(\frac{1}{t^2}\right), \end{aligned}$$

where $R_1(t)$ and $R_2(t)$ are defined in Theorem 1, $\bar{R}_3(t) = \frac{2\beta}{t} E[\hat{e}^T(t-1)(I_d - \frac{\gamma}{t} QW)^T (\mathcal{F}_C(\hat{z}(t-1)) - \frac{1}{\pi_2} \tilde{s}(t))]$.

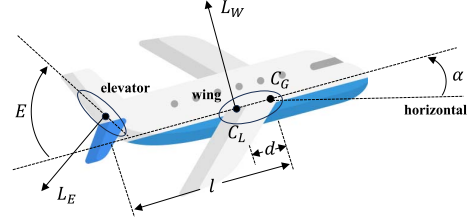


Fig. 2. A schematic diagram of an aircraft.

From Proposition 2, we obtain

$$E\left[\frac{1}{\pi_2} \tilde{s}(t)\right] = E[s(t)] + O(\lambda_p^t) = \mathcal{F}_C((Q \otimes K_2)x(t)) + O(\lambda_p^t).$$

Since $0 < \lambda_p < 1$, this exponential term $O(\lambda_p^t)$ decays rapidly and is negligible compared to $O(\frac{1}{t^2})$. Then, similar to the analysis of $R_3(t)$ in the proof of Theorem 1, we obtain

$$\begin{aligned} \bar{R}_3(t) &= \frac{2\beta}{t} E \left[\hat{e}^T(t-1) \left(I_d - \frac{\gamma}{t} QW \right)^T \left(\mathcal{F}_C(\hat{z}(t-1)) \right. \right. \\ &\quad \left. \left. - \mathcal{F}_C((Q \otimes K_2)x(t)) + O(\lambda_p^t) \right) \right] \\ &\leq -\frac{2\beta f_M}{t} R(t-1) + O\left(\frac{1}{t^2}\right), \end{aligned}$$

which is same as the result (9) of $R_3(t)$.

Following the proof of Theorem 1, we can obtain the conclusion of this theorem, which is omitted here. ■

V. NUMERICAL SIMULATION

This section provides two examples to illustrate theoretical results for the cases of i.i.d. and Markovian packet loss.

Example 1: Consider the altitude consensus control of a multi-aircraft system composed of seven aircraft ([8], [19]), whose communication network is shown in Figure 1.

The schematic diagram of the aircraft is shown in Figure 2. By [19], the dynamics of the i th aircraft is modeled as:

$$\begin{cases} J\ddot{\alpha}_i + b\dot{\alpha}_i + (C_{ZE}l + C_{ZW}d)\dot{\alpha}_i = C_{ZE}lE_i, \\ m\ddot{h}_i = (C_{ZE} + C_{ZW})\alpha_i - C_{ZE}E_i, \end{cases} \quad (10)$$

where h_i is the altitude of i th aircraft controlled by the elevator's rotation E_i , α_i is the rotation angle of i th aircraft.

In this numerical simulation, we set $J = 1, m = 1, b=4, C_{ZE} = 1, C_{ZW} = 5, l = 3$ and $d = 0.2$. Let $x_i = (\alpha_i, \dot{\alpha}_i, h_i, \dot{h}_i)^T$. By adopting the zero-order holder strategy with sampling period $T = 0.5$, the discretization of (10) is obtained as follows:

$$x_i(t+1) = A_d x_i(t) + B_d E_i(t), \quad (11)$$

where

$$A_d = \begin{bmatrix} 0.7358 & 0.1839 & 0 & 0 \\ -0.7358 & 0 & 0 & 0 \\ 0.7073 & 0.0777 & 1 & 0.5 \\ 2.6891 & 0.3964 & 0 & 1 \end{bmatrix}, B_d = \begin{bmatrix} 0.1982 \\ 0.5518 \\ -0.093 \\ -0.2668 \end{bmatrix}.$$

It is easily verified that the linear system described by (11) is controllable. Therefore, it can be transformed into the Brunovsky canonical form defined by (3).

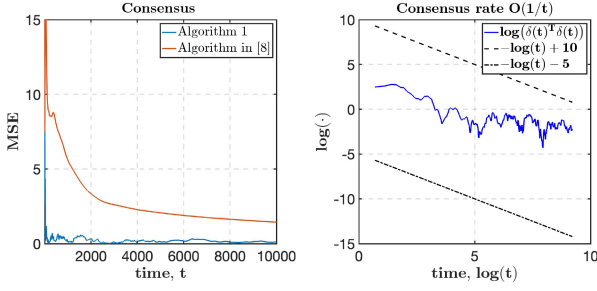


Fig. 3. Trajectories of MSE and log(MSEs) with i.i.d. packet loss.

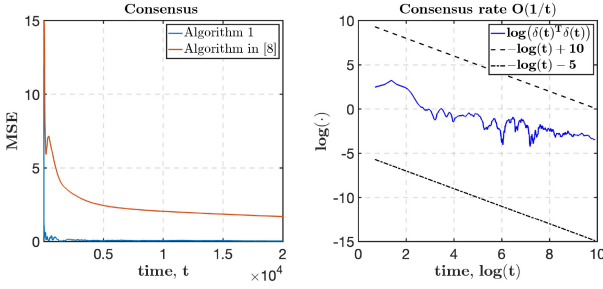


Fig. 4. Trajectories of MSE and log(MSEs) with Markovian packet loss.

In this simulation example, we set the communication noises $d_{ij} \sim \mathcal{N}(0, 16)$ for $(i, j) \in \mathcal{E}$. The packet loss sequence $\{\gamma_{ij}(t)\}_{t \in \mathbb{N}}$ is an i.i.d. Bernoulli process with packet loss probability $p = 0.3$, i.e., $P\{\gamma_{ij}(t) = 1\} = 0.7$ and $P\{\gamma_{ij}(t) = 0\} = 0.3$. Set the thresholds $c_{ij} = -2$.

Then, set the initial altitudes h_i of these seven aircraft are $h_1 = 5, h_2 = 2, h_3 = 4, h_4 = 3, h_5 = 1.5, h_6 = 2.5, h_7 = 1$ and the initial values of $\alpha_i, \dot{\alpha}_i, \dot{h}_i$ are 0. Denote the initial estimates as $\hat{0}_{20}$. Then, we apply the one-bit packet loss consensus algorithm with $\kappa = \frac{1}{1-p} = \frac{10}{7}$, $M = 2$, $\beta = 1500$, $\gamma = 1$, $K_1 = [-0.9224, -0.1825, -0.0000, -0.1788]$, and $K_2 = [3.8734, 0.9054, 0.3575, 0.8772]$.

Figure 3 illustrates trajectories of the consensus mean square errors (MSEs) $\sum_{i=1}^N E[x_i(t) - \bar{x}(t)]^2$ and their logarithm, showing that the agents reach consensus at a rate of $O(\frac{1}{t})$ under i.i.d. packet loss. It also compares Algorithm 1 with the one-bit consensus algorithm without packet loss compensation in [8], demonstrating the effectiveness of Algorithm 1 in dealing with packet loss.

Example 2: Take the same system settings as in Example 1, except the packet loss sequence $\{\gamma_{ij}(t)\}_{t \in \mathbb{N}}$ is an ergodic Markov process with two states $\{0, 1\}$. The transition probability matrix is $P = [1/3, 2/3; 2/7, 5/7]$ with the stationary distribution is $\pi = [0.3, 0.7]$. Then, we apply the one-bit packet loss consensus algorithm with $\kappa = \frac{1}{\pi_2} = \frac{10}{7}$, where $M, \beta, \gamma, K_1, K_2$ are same with Example 1.

From Figure 4, we learn that each agent can reach consensus at the rate of $O(\frac{1}{t})$ with Markovian packet loss.

VI. CONCLUSION

This letter studies the one-bit consensus control problem of MASs with i.i.d. and Markovian packet loss. Quantitative characterizations of the information loss caused by packet loss are provided, establishing the proportional relationships

between one-bit data with and without packet loss in the sense of expectation. Based on these quantitative characterizations, this letter proposes a one-bit packet loss consensus algorithm with a packet loss proportional coefficient designed as the reciprocal of information loss proportion. This letter demonstrates that the proposed algorithm enables MAS to achieve one-bit consensus in the mean square sense at a rate of $O(\frac{1}{t})$ with appropriate step coefficients in both i.i.d. and Markovian packet loss cases. Future research will explore the one-bit consensus with event-triggered communication mechanisms to reduce communication costs further.

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